

MA 403 Real Analysis Homework 4

1. Which of the following subsets of $\mathbb{R}^2$ are compact?
 - (a) the set of all (x, y) such that $x^2 + y^2 = 1$
 - (b) the set of all (x, y) such that $x^2 + y^2 \geq 1$
 - (c) the set of all (x, y) such that $x, y \in \mathbb{Q}$ and $x^2 + y^2 \leq 1$
2. Give an example of a countable open cover F for $\mathbb{Z} \subset \mathbb{R}$ such that F has no finite subcover.
3. Which of the following functions defines a metric on $\mathbb{R}$? (Here $x, y \in \mathbb{R}$.)
 - (a) $d(x, y) = (x - y)^2$
 - (b) $\tilde{d}(x, y) = |x - 2y|$
 - (c) $d^*(x, y) = \frac{|x - y|}{1 + |x - y|}$

Consider the following two functions on $\mathbb{R}^n \times \mathbb{R}^n$:

$$d_1(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|, \quad d_2(x, y) = \sum_{i=1}^n |x_i - y_i|$$

4. Prove that $(\mathbb{R}^n, d_1)$ is a metric space. Prove that $(\mathbb{R}^n, d_2)$ is a metric space.
5. In each of the following metric spaces prove that the ball $B(a; r)$ has the geometric appearance indicated:
 - (a) In $(\mathbb{R}^2, d_1)$, a square with sides parallel to the coordinate axes.
 - (b) In $(\mathbb{R}^2, d_2)$, a square with diagonals parallel to the coordinate axes.